MILCOM 2025 - 2025 IEEE Military Communications Conference (MILCOM) | 979-8-3315-0292-8/25/$31.00 ©2025 IEEE | DOI: 10.1109/MILCOM64451.2025.11310002

Formal Verification of Autonomous Cyber Defense for Abstraction of Critical Network Scenarios

Samy Meziane[1,2], Laurin Holz[1], Tobias Hürten[1], Johannes Loevenich[1],
Thies Möhlenhof[1], and Roberto Rigolin F. Lopes[1]
[1]Secure Communications & Information (SIX), Thales Deutschland, Ditzingen, Germany
[2]École polytechnique, Institut Polytechnique de Paris, Paris, France
Email: {samy.meziane, laurin.holz, tobias.huerten, johannes.loevenich, thies.moehlenhof, roberto.rigolin}@thalesgroup.com

***Abstract*—Autonomous Cyber Defense (ACD) systems require formal guarantees to ensure their reliability and effectiveness in protecting critical network infrastructures. This paper presents a formal verification approach for an abstraction of the Cyber Autonomy Gym for Experimentation (CAGE) Challenge 4 cybersecurity scenario using probabilistic model checking with Probabilistic Symbolic Model Checker (PRISM). We develop a comprehensive multi-zone enterprise network model that captures red-team attack strategies, cross-zone infiltration, and defensive mechanisms. Due to computational complexity and state explosion challenges, we implement a curriculum learning approach, starting with red-only agent models to understand attack behaviors before progressing to adversarial blue-red scenarios. Our methodology aims to prove the feasibility of formal verification for autonomous cyber defense validation, providing probabilistic guarantees for network security properties and establishing a foundation for future multi-agent formal verification frameworks. Our results demonstrate over eight selected scenarios that near-optimal blue policies achieve perfect service degradation prevention while revealing fundamental scalability limits with state spaces growing exponentially from a single-zone model to complete multi-zone scenarios.**

***Index Terms*—Formal Verification, Autonomous Cyber Defense, Probabilistic Model Checking, State Space Explosion**

I. Introduction

The rapid evolution of cyber threats and the increasing sophistication of adversarial tactics have required the development of Autonomous Cyber Defence (ACD) systems capable of real-time threat detection, response, and mitigation. These systems must operate in complex, multi-zone enterprise networks where traditional rule-based security measures prove insufficient against advanced persistent threats [1]. For example, the Cyber Autonomy Gym for Experimentation (CAGE) Challenge 4 [2] presents a realistic enterprise network environment where a team of autonomous agents must defend against sophisticated red team attacks across multiple security zones with varying trust levels and connectivity constraints. This is particularly relevant for military applications, as it models the type of multi-zone enterprise networks commonly found in defense organizations where operational technology systems, classified repositories, and administrative networks must be protected against sophisticated nation-state adversaries while maintaining mission-critical operational capabilities.

Current approaches to validate ACD agents rely on simulation-based testing, which cannot provide formal guarantees for critical infrastructure deployment. While Multi Agent Reinforcement Learning (MARL) techniques, particularly Centralized Training with Decentralized Execution (CTDE) [3], show promise in cyber defense policies, the black-box nature of neural networks makes verification challenging. To address these challenges, this paper develops a formal verification framework using the Probabilistic Symbolic Model Checker (PRISM) [4] for the CAGE Challenge 4 scenario, employing a curriculum learning approach to manage computational complexity.

The main contributions of this work are: (1) the development of a formal model of the CAGE Challenge 4 network environment suitable for probabilistic model checking; (2) a curriculum learning approach that addresses computational complexity challenges through progressive model abstraction and refinement; (3) formal verification of critical security properties including attack success probabilities, cross-zone infiltration rates, and defensive effectiveness; (4) the current limits of formal verification to the state space explosion problem; and (5) establishment of a foundation for future research in multi-agent formal verification for cybersecurity.

The remainder of this paper is organized as follows. Section II reviews related work in formal verification for cybersecurity and autonomous defense systems. Section III describes our formal verification methodology, including the challenge scenario, the curriculum learning approach, and abstraction techniques. Section IV presents our results and discusses the implications for ACD verification. Section V concludes with a summary of our contributions and outlines future research directions.

II. Background & Related Work

Formal verification of ACD systems is highly relevant to North Atlantic Treaty Organization (NATO), as it directly addresses the Alliance's strategic priority of ensuring trust, robustness, and interoperability in coalition cyber defence operations. NATO networks, particularly those implementing Federated Mission Networking (FMN) and Protected Core Networking (PCN), must withstand advanced persistent threats

Authorized licensed use limited to: Ecole Nationale des Ponts et Chaussées. Downloaded on April 10,2026 at 12:30:13 UTC from IEEE Xplore. Restrictions apply.

from nation-state adversaries while preserving mission-critical capabilities. Current simulation-based testing provides limited assurance, whereas formal verification offers mathematically rigorous guarantees of security properties under adversarial conditions. This work supports NATO's goals of improving resilience, enabling human–machine teaming, and improving confidence in the deployment of ACD agents across multinational mission environments by demonstrating scalable methods to verify defensive effectiveness and containment in multi-zone enterprise networks.

Recent research in formal verification for the cybersecurity domain has demonstrated the potential to increase the robustness and trustworthiness of Autonomous Cyber Defence (ACD) systems for critical network infrastructure used by the military [5]. Early work focused primarily on static analysis and bounded model checking for security protocols and cryptographic systems. However, multi-agent cyber defense systems have introduced new challenges that traditional formal methods struggle to address. For example, the solution in [6] complements these methods by introducing A Comprehensive Tool for Reinforcement Learning and Model Checking (COOL-MC), a tool capable of verifying Deep Reinforcement Learning (DRL) agents during the training phase, and thus enabling formal guarantees at any stage of the model lifecycle.

In [7] the authors introduce a mathematical model for cyber defense scenarios using CAGE Challenge 2 [8]. Their approach demonstrated the feasibility of using probabilistic model checking for analyzing cyber defense strategies in simplified network environments. The authors developed a mathematical model that captured basic attack-defense interactions, focusing on single-zone scenarios with limited agent capabilities.

Cybersecurity systems are typically defined over indefinite continuous spaces, which is at odds with a fundamental requirement for providing formal guarantees: the construction of finite abstractions that preserve essential properties while remaining computationally tractable. The authors in [9] demonstrate this principle in their work on DRL and formal verification, where they develop interval Markov Decision Process (MDP) abstractions that yield probabilistic guarantees on policy execution. Their approach illustrates the critical trade-off between abstraction fidelity and computational feasibility. The authors show that systematic abstraction refinement can achieve the finite state spaces necessary for exhaustive model checking while preserving the probabilistic properties required for meaningful system analysis.

The paradigm of CTDE [10] represents a frequent choice for cyber defense scenarios where ACD agents must learn coordinated defensive strategies across multiple network zones while maintaining operational autonomy during deployment. The centralized training phase enables agents to develop sophisticated coordination mechanisms and shared situational awareness that would be difficult to achieve through purely decentralized learning approaches. In [11] the authors introduce a MARL framework for cyber defense scenarios. Their CTDE implementation allows agents to share global state information during training while operating entirely on local information during execution, addressing the partial observability challenges inherent to the domain [3]. Yet, these techniques are limited by the black-box nature of neural networks and the state space explosion problem that makes them difficult to verify.

Recent advances in adversarial machine learning have highlighted additional challenges for autonomous cyber defense systems. The investigation in [12] demonstrated that even well-trained agents can be vulnerable to carefully crafted adversarial samples that exploit the black-box nature of neural network policies. This underlines the importance of formal verification approaches that allow mathematical guarantees about system behavior even in the presence of sophisticated adversarial policies in order to trust MARL-based ACD solutions in the future.

Challenges related to computational complexity associated with formal verification of large-scale systems have been addressed through various abstraction techniques. Counterexample-guided abstraction refinement (CEGAR) [13] and predicate abstraction have shown promise in reducing state space complexity for software verification, but their application to MARL remains largely unexplored. Similarly, compositional verification approaches that decompose large systems into smaller, more manageable components have been applied to distributed systems verification, but require careful consideration of agents deployed in cyber defense environments.

There remains a research gap in the context of formal guarantees for MARL-based ACD. Our work is motivated by the limitations in the black-box nature inherent to neural networks, and the computational intractability of exhaustive state space exploration for complex MARL systems. We motivate a curriculum learning approach paired with a progressive abstraction methodology, aiming to address these fundamental challenges while preserving the completeness guarantees that formal verification provides.

III. Methodology

This work addresses the fundamental challenge of applying verification techniques to multi-zone cybersecurity scenarios through progressive model abstraction and a curriculum learning approach that gradually increases model complexity.

A. Critical Network Scenario

The CAGE Challenge 4 scenario [2] models a realistic, non-deterministic and non-stationary multi-zone enterprise network consisting of six distinct zones connected via untrusted infrastructure: Contractor (external entry point), Headquarters (administrative hub), two Restricted zones (sensitive data), and two operational zones (mission-critical systems), with controlled inter-zone connectivity following defense-in-depth principles. The environment features three types of agents operating under *partial observability*: Red team attackers capable of phishing, reconnaissance, exploitation, lateral

Authorized licensed use limited to: Ecole Nationale des Ponts et Chaussées. Downloaded on April 10,2026 at 12:30:13 UTC from IEEE Xplore. Restrictions apply.

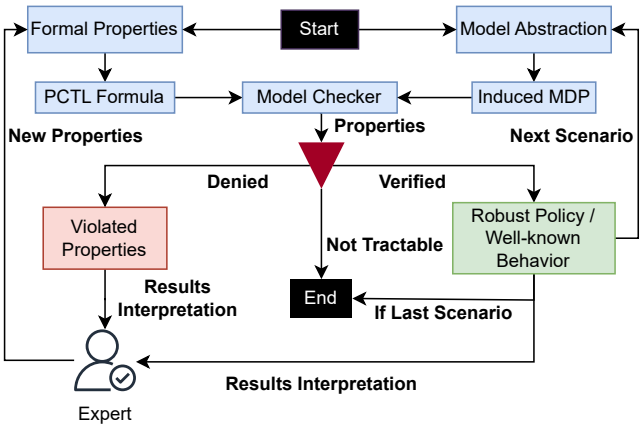

Fig. 1: Model Checking Workflow Using Curriculum Learning.

movement, and service degradation through impact actions and data exfiltration; blue team defenders performing monitoring, analysis, response (including decoy deployment and machine restoration), and recovery actions; and passive green agents representing hosts that require service operability while potentially falling victim to phishing attacks.

Each zone contains heterogeneous machines (servers, workstations, infrastructure devices) with varying security postures, creating complex attack surfaces where red agents must navigate incomplete knowledge of the network's topology and defense measures, while blue agents face uncertainty about red team presence across their respective zones and attack progression. This complex mixed, inhomogeneous multi-agent environment with incomplete information and stochastic transitions can be modeled as a Partially Observable Markov Decision Process (POMDP).

B. Curriculum Learning for Cyber Defense Analysis

This section introduces a formal verification workflow in Fig. 1 following an iterative process that transforms abstracted scenarios and formal security properties into verifiable mathematical models. This begins with abstracted scenarios derived from the curriculum learning approach, which are converted into induced MDPs through the PRISM modeling language. Simultaneously, the properties are mathematically formulated into Probabilistic Computation Tree Logic (PCTL) expressions that capture the specific behavioral guarantees we seek to verify. The model checker then processes both the MDP and PCTL formulations to determine verification outcomes. Depending on the results, it reveals a sound comprehension of the agents' behavior or a need for refinement of the abstraction.

We employ a curriculum learning approach, similar to [10], that begins with simplified abstractions and progressively increases complexity to approach realistic scenarios: (1) one zone with only one red agent, (2) one zone with one red and one blue agent, (3) two zones with two red agents, (4) two zones with two agents each, (5) four zones with two agents each, (6) all the zones with only the red team, and (7) the whole scenario. This methodology is inspired by educational principles where complex concepts are learned by mastering simpler building blocks first. In our context, curriculum learning enables us to derive a comprehensive understanding of agent behaviors at each step to guarantee security-relevant properties by isolating specific interaction patterns before introducing additional complexity.

The curriculum learning principle provides several advantages for formal verification of cyber defense scenarios. First, it allows analysis of fundamental adversarial dynamics without the computational burden of modeling all system components simultaneously. Second, it enables validation of modeling assumptions and verification techniques on tractable problems before scaling to complex scenarios. Third, it provides interpretable insights into agent behaviors at each level of complexity, addressing the black-box problem inherent in neural networks by using near-optimal mathematical policies that can be formally analyzed.

Acknowledging the fundamental trade-off between fidelity and verification tractability, we model the adversarial interactions as a turn-based scenario to enable systematic formal analysis. This formulation provides several analytical advantages: it eliminates race conditions and temporal uncertainties that would significantly complicate the state space while taking into account the discrete time POMDP, it enables precise analysis of action sequences and their probabilistic outcomes, and it allows exhaustive exploration of decision points for both blue and red agents. Each turn represents a decision point where agents can observe the current state, select actions based on their policies, and affect the transition to new states based on the probabilistic outcomes of their actions. This formulation preserves the core adversarial dynamics while enabling the exhaustive state space exploration required for formal guarantees.

C. Model Abstraction and State Space Challenges

The environment state consists of machine states, agent position, network connectivity, and security configurations across all zones. Machine states capture the current security posture of individual systems, including running services, user sessions, installed software, and evidence of compromise. Agent states track the current positions, capabilities, and knowledge of both red and blue agents across the network.

Even with simplified models, the state space grows exponentially with the number of machines, zones, and agent capabilities, creating trillions of states that challenge the computational limits of current model checking techniques. This state explosion arises from the multiplicative nature of state component combinations, i.e. each machine can exist in multiple compromise states, agents can be positioned in various locations, and defenders can deploy countermeasures.

To address these challenges, we structure our methodology around two primary objectives that balance analytical depth with computational feasibility:

- *Objective 1*: Single/Dual-Zone Blue-Red Adversarial Dynamics. This objective analyzes adversarial dynamics

Authorized licensed use limited to: Ecole Nationale des Ponts et Chaussées. Downloaded on April 10,2026 at 12:30:13 UTC from IEEE Xplore. Restrictions apply.

between red and blue agents within constrained environments. By replacing neural network-based policies with near-optimal mathematical policies, we circumvent black-box limitations and enable formal analysis of blue team detection, response strategies, and red team persistence patterns under defensive pressure.

- *Objective 2*: Red Team Behavior Analysis. This objective examines red team attack progression across all network zones to understand lateral movement patterns and multi-stage campaigns. The all-zones red-only model provides formal specifications for hybrid verification approaches, where a Python framework queries PRISM-based red actions, provides observations to blue MARL agents, and continues simulation, combining formal guarantees with learned policies.

D. State Space Reduction Techniques

To mitigate the state explosion problem, we explore several reduction techniques that exploit the structural properties of cyber defense scenarios. Leveraging *symmetry* in the network, i.e. using that many network hosts within and between zones share an identical architecture, we represent machines with equivalent security states as a single abstract machine. Moreover, *Zone-level parallelism* enables the exploitation of the relative independence of activities in different network zones when direct connectivity does not exist, allowing compositional analysis techniques.

We can further reduce the branching factor in the state space by *abstracting the Action Group* of offensive and defensive actions that produce equivalent state transitions. Finally, *Temporal Abstraction* enables bounded model checking approaches by providing guarantees for properties with a real-time horizon, instead of verifying the entire actions space at every state. These reduction techniques ensure the preservation of essential security properties while achieving computational tractability. While we have not yet fully implemented all these advanced reduction techniques, our curriculum learning approach still enables verification of key properties through progressive complexity management.

E. PRISM Model Implementation

We implement our models using the PRISM model checker and Fig. 1 methodology, which provides a high-level modeling language for specifying probabilistic finite state machines and MDPs. The PRISM framework enables both probabilistic model checking and discrete-time simulation, supporting comprehensive analysis of stochastic cybersecurity scenarios. Each zone is modeled as a collection of machines with explicit state variables capturing security-relevant properties. It includes operational status, compromise indicators, running services, and agent presence. State transitions model the effect of agent actions on the machine security posture and the propagation of compromise throughout the network.

Agent behaviors are modeled through action-dependent state transitions with associated probabilities reflecting the uncertainties inherent to cyber operations, as well as the strategy of the red team. Attack success probabilities depend on system vulnerabilities, defense countermeasures, and environmental factors. Defense actions similarly exhibit probabilistic outcomes based on detection capabilities and response effectiveness. Inter-zone connectivity is explicitly modeled through conditional transitions that respect network topology constraints. Each zone has a dormant red agent that can be activated only when lateral movement actions succeed and when appropriate network connections exist. This approach ensures that formal verification results accurately reflect the intended network architecture.

We develop comprehensive property specifications using PCTL to capture security-relevant behaviors and requirements. Properties are organized into categories addressing different aspects of cybersecurity analysis, where all properties include explicit time bounds to ensure verification termination and computational feasibility.

Reachability properties including *persistence and escalation properties* quantify the probability of attackers reaching critical assets or zones within specified time bounds, or maintaining access and privileges even under defensive pressure. These properties directly address fundamental security questions about attack success likelihood and defensive effectiveness. *Adversarial dynamics properties* capture the complex interactions between blue and red agents, such as detection patterns, response effectiveness, and adaptation strategies. *Defensive posture properties* assess blue team effectiveness in prevention, detection, and response across different threat scenarios. Finally, *cross-zone containment properties* analyze the effectiveness of defensive measures in preventing lateral movement and zone infiltration, which is critical for maintaining overall network security.

IV. Results & Discussion

Our formal verification approach demonstrates the effectiveness of curriculum learning and progressive abstraction for analyzing complex cybersecurity scenarios while providing quantitative insights into adversarial dynamics. The results validate key hypotheses about defensive effectiveness and reveal important computational challenges that inform the future development of hybrid verification frameworks.

A. Model Implementation and State Space Analysis

We fully implemented eight distinct PRISM models covering progressively complex scenarios from single-zone operations to the complete six-zone CAGE Challenge 4 environment following the curriculum learning approach. The *single-zone models* (A, C, see Table I) serve as baselines to understand isolated agents dynamics, the *two-zones models* (B, D, E) introduce cross-zone connectivity and the *multi-zone models* (F, G, H) represent the full scenario for further studies. Fig. 2 plots the number of states (y-axis) for the 8 scenarios (x-axis) to demonstrate the fundamental computational challenge in formal verification of cyber defense

Authorized licensed use limited to: Ecole Nationale des Ponts et Chaussées. Downloaded on April 10,2026 at 12:30:13 UTC from IEEE Xplore. Restrictions apply.

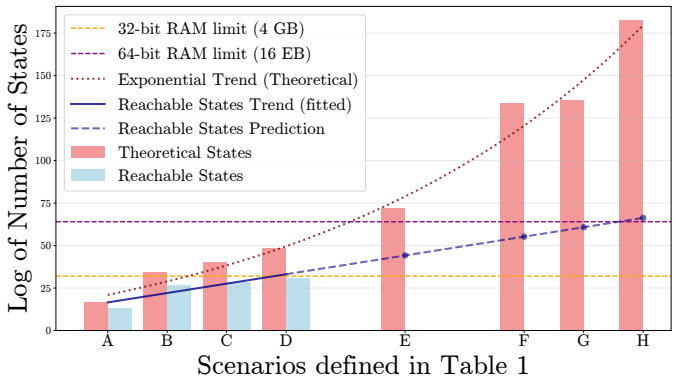

Fig. 2: State Space Complexity Across Model Variants.

scenarios: exponential growth in theoretical states from 2^{16} (single-zone) to 2^{183} (whole scenario), far beyond current verification capabilities. Table I shows reachable states follow a similar reduced trend due to action constraints, with transitions also representing major memory storage issues, making every scenario with more than ∼50 billion states intractable. Blue agent interactions dramatically increase complexity, making only simplified two-zone models computationally tractable.

Notably, the reduction factor between theoretical and reachable states increases substantially with model complexity (from 90.4% to 99.9994%), ranging from modest reductions in single-zone scenarios to several orders of magnitude for multi-zone models, demonstrating that actions constraints and state dependencies provide significant mitigation of the exponential growth challenge. This remains insufficient now for large-scale scenarios, but the predicted trend in Fig. 2 gives promising possible results if the reachable states for these scenarios really follow this trend.

B. Computational Performance and Scalability Limits

Our methodology successfully addresses computational challenges for single and multi-zone scenarios but reveals fundamental scalability limits inherent to formal verification and model checking that necessitate hybrid approaches for realistic enterprise networks. State space analysis shows that pure formal verification becomes computationally intractable beyond two zones with comprehensive agent interactions.

The single-zone models can be completely verified within hours on medium-range computing power, enabling comprehensive property analysis albeit still lengthy. Two-zone models require simplification and several hours for comprehensive property verification but remain key models to understand cross-zone dynamics. The six-zone, four-zone and the complete two-zone models present prohibitive computational challenges, with full state space exploration needing extensive memory and computing power.

These scalability limitations show the purpose of improving formal verification processes to be able to entrust enterprise networks to MARL-based agents in the near future. These limits also validate our dual-objective approach: detailed formal verification for single-zone adversarial dynamics, and multi-zone scenarios (with only red agents or fewer machines) for cross-zone dynamics comprehension and allow hybrid integration of frameworks that leverage formal baselines with simulation-based analysis for practical tractability.

TABLE I: State Space Complexity Analysis.

Scenario	Theoretical	Reachable	Reduction	Transitions
Single Zone Scenarios				
A: 1 Zone Red-Only	100,000	9,604	90.4%	122,829
C: 1 Zone Blue vs Red	1.558e12	347,035,825	99.98%	1,771,481,620
Two Zone Scenarios				
B: 2 Zones Red-Only	2e10	104,683,600	99.48%	2,696,860,908
D: 2 Zones 6 Machines	3.740e14	2,268,750,719	99.9994%	8,900,249,259
E: 2 Zones Blue vs Red	4.847e21	Intractable	N/A	N/A
Multi-Zone Scenarios				
F: 4 Zones Blue vs Red	2.345e40	Intractable	N/A	N/A
G: All Zones Red-Only	6.872e40	Intractable	N/A	N/A
H: All Zones Blue vs Red	1.003e55	Intractable	N/A	N/A

C. Verified Properties: Strategic Implications and Insights

Table II lists formal results for the first objective of our methodology: characterizing the adversarial dynamics in one and two zones. These results quantify the impact of defensive measures on attack success while revealing the attack power of the red team when the blue team is only choosing the no-op action *Sleep*. These results also show the impact of near-optimal defensive policies on attack success while revealing complex strategies for interactions between red and blue agents. Even with the high number of states and transitions, verification remains feasible for the first four scenarios of Fig. 2 but the verification time can grow to more than ten hours for some selected properties, whereas other properties remain very hard to verify, highlighting the need to further simplify the scenario for the two-zones dynamics.

Perfect Blue Defensive Success. The near-optimal blue policy achieves complete prevention of any data exfiltration on any machine and achieves very good results preventing Red from reaching root access by almost every time detecting the ongoing attack. This shows the theoretical upper bound of defensive effectiveness and the big effect on red attack power, where these probabilities without blue defense were 100%.

Adversarial Persistence vs. Impact Prevention. While near-optimal blue policies achieve perfect impact prevention, sophisticated adversaries maintain significant operational capabilities, highlighting the ongoing nature of cyber defense rather than achieving permanent security states. This effect is also evident in the decoy effectiveness: while decoy deployment succeeds with probability 1, the persistent decoy effectiveness drops to a minimum probability of 0, indicating that deception measures and defensive efforts require continuous renewal to remain effective.

Cross-zone propagation analysis reveals security vulnerabilities in multi-zone networks. Our formal verification demonstrates that lateral movement always succeeds eventually. Notably, Blue team manages to remove the Red agent from the Operational zone with probability 1, while Restricted zone security becomes virtually impossible (probability 1.23e-13)

Authorized licensed use limited to: Ecole Nationale des Ponts et Chaussées. Downloaded on April 10,2026 at 12:30:13 UTC from IEEE Xplore. Restrictions apply.

TABLE II: Verified Properties: Blue vs. Red Scenarios.

Property Category	Property Description	Probability	Verification
Attack Success	Red gains user access eventually	0.999	✓
	Red achieves full impact on any machine of the two zones	0.000	✓
	Red bypasses deception eventually	1.000	✓
Blue Defense Effectiveness	Blue maintains decoys on machines	1.000	✓
	Blue detects the ongoing attack	0.999	✓
	Blue is able to prevent any root access	0.831	✓
Adversarial Dynamics	If Red gains user/root access over a host, then within one turn, Blue chooses Remove/Restore actions	1.000	✓
Cross-Zone Dynamics	Red infiltrates Operational Zone eventually	1.000	✓
	Restricted Zone remains completely secure	1.23e-13	✓
	Blue removes Red from the Operational Zone eventually	1.000	✓

due to initial red team positioning. These results quantify the asymmetric nature of zone compromise: once initial access is established, zone-wide compromise becomes likely, but cross-zone propagation faces significant barriers that can be exploited for containment strategies.

Defense-in-Depth Validation. These results provide quantitative validation of defense-in-depth principles, showing that layered defensive capabilities (detection $\rightarrow$ analysis $\rightarrow$ response) can achieve perfect security outcomes under mathematical optimization. The near-optimal policy successfully passes all critical security threshold tests, showing the effectiveness of this optimization while being computationally tractable.

Implication for CAGE Challenge 4 Integration. These results provide essential baselines for developing the hybrid verification framework targeting complete CAGE Challenge 4 scenarios. The two-zones dynamics results are promising here because the Red team needs to cross into a zone, where the Blue team already is, to access the network. The quantified probabilities establish performance thresholds that MARL-based blue agents must achieve to maintain security equivalence with the formally verified policies. As these agents are often trained with near-optimal policy guidance, this also shows the relevance of training such agents for multi-zone environments where the attack surface and complexity increase substantially for a human being to protect these networks.

D. Model Validation and Real World Limitations

Our model captures essential aspects of cybersecurity scenarios while necessarily simplifying complex real-world systems. Validation through property demonstrates that the models exhibit expected behaviors for fundamental security scenarios. Attack progression patterns match expected cyber kill chain dynamics, with attackers requiring reconnaissance phases before successful exploitation and lateral movement. Defensive interventions appropriately disrupt attack progression and reduce success probabilities. Inter-zone movement respects network topology with the sole exception of non-existing firewalls and requires appropriate attack capabilities.

V. Conclusion

This work demonstrates the successful application of formal verification techniques to complex cybersecurity scenarios through a curriculum learning approach and progressive model abstraction with a comprehensive PRISM-based framework for the CAGE Challenge 4 scenario. Our methodology addresses fundamental computational challenges while providing quantitative insights into adversarial dynamics that are difficult to obtain without simplification measures. Nevertheless, we verified key properties for the CAGE Challenge 4 scenario and the near-optimal blue agent, demonstrating its ability to defend one zone effectively despite the action space of the red team and to maintain zone connectivity despite attacker persistence. The progressive abstraction approach enables systematic analysis of security properties at increasing levels of complexity while revealing the computational boundaries of formal verification approaches. Our results establish formal verification as a viable complement to existing cybersecurity analysis methods while revealing critical scalability challenges inherent to multi-zone networks and multi-agent verification.

References

[1] J. Loevenich, E. Adler, T. Hürten, and R. R. F. Lopes, "Design and evaluation of an Autonomous Cyber Defence agent using DRL and an augmented LLM," *Computer Networks*, vol. 262, p. 111162, 2025.

[2] T. C. W. Group. (2023) *TTCP CAGE Challenge 4* [Online]. Available: https://github.com/cage-challenge/cage-challenge-4.

[3] G. Papoudakis, F. Christianos, and S. V. Albrecht, "Agent Modelling under Partial Observability for Deep Reinforcement Learning," in *35th Conference on Neural Information Processing Systems*, 2021.

[4] M. Kwiatkowska, G. Norman, and D. Parker, "PRISM 4.0: Verification of Probabilistic Real-Time Systems," in *Gopalakrishnan, G., Qadeer, S. (eds) Computer Aided Verification. CAV 2011. Lecture Notes in Computer Science*, vol. 6806, 2011.

[5] L. Holz, J. Loevenich, and R. R. F. Lopes, "Towards Robust Autonomous Cyber Defence Agents Using Hybrid AI Models," in *2025 IEEE 11th International Conference on Network Softwarization (NetSoft)*, 2025, pp. 269–272.

[6] D. Gross, N. Jansen, S. Junges, and G. A. Pérez, "Cool-mc: A comprehensive tool for reinforcement learning and model checking," in *Dependable Software Engineering. Theories, Tools, and Applications*, W. Dong and J.-P. Talpin, Eds. Cham: Springer Nature Switzerland, 2022, pp. 41–49.

[7] K. Hammar, N. Dhir, and R. Stadler, "Optimal Defender Strategies for CAGE-2 using Causal Modeling and Tree Search," *arXiv*, pp. 1–17, 2024.

[8] M. Kiely, D. Bowman, M. Standen, and C. Moir, "On autonomous agents in a cyber defence environment," *arXiv preprint arXiv:2309.07388*, 2023.

[9] E. Bacci and D. Parker, "Verified Probabilistic Policies for Deep Reinforcement Learning," in *Proc. 14th International Symposium NASA Formal Methods (NFM'22)*, vol. 13260, 2022, pp. 193–212.

[10] T. Hürten, J. F. Loevenich, F. Spelter, E. Adler, J. Braun, L. Moxon, Y. Gourlet, T. Lefeuvre, and R. R. F. Lopes, "Hierarchical Multi-Agent Reinforcement Learning for Autonomous Cyber Defense in Coalition Networks," in *MILCOM 2024 - 2024 IEEE Military Communications Conference (MILCOM)*, 2024, pp. 176–181.

[11] J. Wiebe, R. A. Mallah, and L. Li, "Learning cyber defence tactics from scratch with multi-agent reinforcement learning," 2023. [Online]. Available: https://arxiv.org/abs/2310.05939

[12] D. Khachaturov, I. Shumailov, Y. Zhao, N. Papernot, and R. Anderson, "Markpainting: Adversarial machine learning meet inpainting," in *38th International Conference on Machine Learning*, 2021.

[13] E.Clarke, O.Grumberg, S.Jha, Y.Lu, and H.Veith, "Counterexample-guided abstraction refinement," in *Emerson, E.A, Sistla, A.P. (eds) Computer Aided Verification. CAV 2000. Lecture Notes in Computer Science*, vol. 1855, 2000.

Authorized licensed use limited to: Ecole Nationale des Ponts et Chaussées. Downloaded on April 10,2026 at 12:30:13 UTC from IEEE Xplore. Restrictions apply.